SQL(Basic)

Assignment

Problem Statement:

You have been assigned to work as a Database Developer for an online bookstore. The bookstore maintains information about books and their authors in a relational database. The business team requires SQL queries to retrieve, analyze, and manage the data efficiently.

The objective of this assignment is to demonstrate your understanding of SQL fundamentals, including data retrieval, filtering, sorting, grouping, joins, data manipulation, and table creation using SQL.

Create the required database tables, populate them with sample data, and write SQL queries to complete the tasks listed below.

Database Schema

Create the following tables:

Authors

Column	Data Type
author_id	INT
author_name	VARCHAR(100)
country	VARCHAR(50)

Books

Column	Data Type
book_id	INT
title	VARCHAR(100)
author_id	INT
category	VARCHAR(50)
price	DECIMAL(10,2)
published_year	INT

Populate both tables with:

- At least 5 authors

- At least 10 books
- Books from different categories
- Authors from different countries

Tasks:

Section 1: Basic Queries
- Task 1: Display all books.
- Task 2: Display only the book title and price.
- Task 3: Display the book title as Book Name and price as Book Price using column aliases.
- Task 4: Display all unique book categories.

Section 2: Filtering Data
- Task 5: Display books priced above ₹700.
- Task 6: Display books priced between ₹500 and ₹800.
- Task 7: Display books that belong to the Programming category.
- Task 8: Display books whose titles start with the letter S.
- Task 9: Display books published after 2020.

Section 3: Sorting and Limiting Results
- Task 10: Display books sorted by price in descending order.
- Task 11: Display the top 3 most expensive books.

Section 4: Aggregate Functions
- Task 12: Find:
  Total number of books
  Total price of all books
  Average book price
  Highest book price
  Lowest book price

Section 5: GROUP BY and HAVING
- Task 13: Display the average book price for each category.
- Task 14: Display only those categories whose average book price is greater than ₹650.

Section 6: Joins
- Task 15: Display each book along with its author's name using an INNER JOIN.
- Task 16: Display all authors along with the books they have written using a LEFT JOIN.
- Task 17: Display all books along with their author details using a RIGHT JOIN.

Section 7: Data Manipulation
- Task 18: Insert a new book into the Books table.
- Task 19: Update the price of the book you inserted.
- Task 20: Delete the book record that you inserted.

Section 8: Creating Tables and Constraints

Task 21: Create a new table named Publishers with the following columns:

publisher_id
publisher_name
country
established_year

Apply the following constraints:

PRIMARY KEY
NOT NULL
UNIQUE

DEFAULT

Deliverables:

Submit the following:

A single .sql file containing:

- CREATE TABLE statements
- Sample INSERT statements
- SQL queries for all tasks
- Organize the SQL script in the same order as the tasks mentioned in the assignment, using appropriate comments (for example, -- Task 1, -- Task 2, etc.) to make the script easy to review.

Upload the .sql file to a GitHub repository.

Share:

- GitHub Repository link
- Pull Request (PR) link
- Branch name used for implementation

A README.md containing:

- Assignment overview
- Database schema
- Steps to execute the SQL script

Screenshots showing the successful execution of:

- Aggregate Functions (Task 12)
- INNER JOIN (Task 15)
- CREATE TABLE (Task 21)

Share daily status updates through email during the assignment.

[End]